A Simple Proof of Pi Value and Euler's Identity

Chinnaraji Annamalai
School of Management, Indian Institute of Technology, Kharagpur, India
Email: anna@iitkgp.ac.in
https://orcid.org/0000-0002-0992-2584

Abstract: This paper presents a technique to prove the Euler's identity and value of Pi.

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Euler's Identity and Pi-value

Euler's identity [1] is that $e^{i\pi} + 1 = 0$.

Proof. Let us prove the Euler's identity using De Moivre's formula.

$$\text{De Moivre's formula: } e^{xi} = \cos x + i \sin x. \quad (1)$$

By substituting $x = \pi$ in (1), we obtain

$$e^{\pi i} = \cos \pi + i \sin \pi$$

$$= \cos\left(\frac{\pi}{2} + \frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2} + \frac{\pi}{2}\right)$$

$$= -\sin\frac{\pi}{2} + i \cos\frac{\pi}{2}$$

$$e^{\pi i} = -1 + i0 = -1$$

$$\therefore e^{i\pi} + 1 = 0.$$

Hence, the Euler's identity is proved.

Pi-value

$$\text{The value of Pi: } \pi = \frac{22}{7}.$$

Proof. Let prove the value of Pi [2] using the circumference of a circle.

circumference of a circle $= 2\pi r$, where r is radius and 2r is diameter.

i. e. Circumference $= \pi \times$ Diameter.

$$\text{Thus, } \pi = \frac{\text{circumference}}{\text{diameter}}.$$

If the circumference of a circle is 14 units, then diameter is 7 units and vice-versa, where unit denotes millimeter(mm), centimeter(cm), meter(m), etc.

$$\therefore \pi = \frac{\text{circumference}}{\text{diameter}} = \frac{22}{7}.$$

Hence, the value of Pi is proved.

References

[1] Yue-Xian, L. (2017) Lecture 5: Complex Numbers and Euler's Formula. *University of British Columbia, Vancouver*. https://personal.math.ubc.ca/~yxli/m152_L5_2017.pdf.

[2] Annamalai, C. (2009) Annamalai;s Google website. https://sites.google.com/view/chinnaraji-annamalai